

3(a)	Is the BG on non-judicial stamp paper of appropriate value?		
3(b)	Is the date of sale of non-judicial stamp paper shown on the BG and the stamp paper is issued not more than Six months prior to the date of execution of BG?		
4(a)	Are the factual details such as Bid specification No., LOA No. contract price, etc, correct?		
4(b)	Whether Overwriting /cutting, if any on the BG, authenticated under signature & seal of executants?		
5	Is the amount and validity of BG is in line with contract provisions?		
6	Whether the BG has been issued by a Nationalized bank / Non- Nationalized Bank acceptable to Buyer /Scheduled Bank in India (the applicability of the bank should be in line with the provisions of bidding documents)?		

Dated the *[Insert date of the month]* day of *[Insert month, year]*
at *[Insert place]*.

Signature {(of Bidder's authorized Bid Signatory (ies))[#] {In full and initials}}:

Full name: {insert full name of authorized Bid Signatory }

Title: {insert title/position of authorized Bid Signatory }

Name of Bidder (Sole Bidder's name or Consortium/ JV's name, if applicable):

Capacity: {insert the person's capacity to sign for the Bidder}

Address: {insert the authorized Bid Signatory's address}

Phone/fax: {insert the authorized Bid Signatory's phone and fax number, if applicable}

Email: {insert the authorized Bid Signatory's email address} _____

(Common Seal).....

Form 20**(Appendix to Technical Part of the Bid)****Format of Additional Information**

NIT/RFB No.:

*[insert details]*Package Name/ Contract Title: *[insert details]*

Page _____ of _____ pages

Bidder's Name and Address:

To:

The Chief Engineer (Spl. Projects),

MAHARASHTRA STATE ELECTRICITY DISTRIBUTION COMPANY LIMITED

Plot no. G-9, 5th floor, Prakashgad, Anant Kanekar Marg, Bandra (East), Mumbai - 400051*{In case of JV bidder, mention name**and address of all the Joint Venture members}***(Additional Information)**

Dear Sir/ Madam,

In support of the additional information required as per the Bidding Documents, we furnish herewith our data/details/documents etc., along-with other information, as follows (the stipulations have been reproduced in italics for ready reference):

1.0 The Bidder shall furnish

A certificate from their Banker(s) (as per prescribed formats in Form 16, Part -3, Section-8: Contract Forms) indicating various fund based/non fund based limits sanctioned to the Bidder and the extent of utilization as on date. Such certificate should have been issued not earlier than three months prior to the date of bid opening.

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Wherever necessary the Employer may make queries with the Bidders' Bankers.
[Reference Part -1, Section 2]

- 1.1 In accordance with 1.0, certificate(s) from banker as per requisite format, indicating various fund based/non fund based limits sanctioned to the bidder or each member of the joint venture and the extent of utilization as on date is/are enclosed, as per the following details:

Name of the Bidder/partner of Joint Venture	
Name of the Banker by whom certificate issued	
Date of certificate (should not be earlier than 3 months prior to date of bid opening)	
Whether fund based/non fund based limits are indicated in the certificate	
Whether extent of utilization is indicated in the certificate	

- 1.2 *The Bidder should accordingly also provide the following information/documents (**In case of JV bidders, information should be provided separately for all the Partners of JV in the given format**):*

- (i) Details of Banker:

Name of Banker	
Address of Banker	
Telephone No.	
Contact Name and Title	

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Fax No.	_____
E-mail ID	_____

(ii) As per para 1.0, Authorization Letter(s) from the bidder (in case of JV bidder, from all the partners) addressed to the Banker(s), authorizing **The Chief Engineer (Spl. Projects) MAHARASHTRA STATE ELECTRICITY DISTRIBUTION COMPANY LIMITED**

to seek queries about the bidder with the Banker(s) and advising the Banker(s) to reply the same promptly, is/are enclosed as per following details:

Sl. No.	Letter Ref.	Date	Addressed to (name of the Bank)

2.0 OTHER INFORMATION

2.1 Current Contract Commitments of works in progress

Bidders (individual firms or each partners of JV) should provide information on their current commitments on all contracts that have been awarded, or for which a letter of intent or acceptance has been received, or for contracts approaching completion, but for which an unqualified, full completion certificate has yet to be issued.

Details of Contract	Value of outstanding work (Rs.)	Estimated completion date

2.2 Financial Data :

(In Rs. Lakhs)

	Actual (previous five years)					Projection for next five years				
1. Total Assets										
2. Current Assets										
3. Total Liability										
4. Current Liability										
5. Profit before taxes										
6. Profit after taxes										

3. The information/documentation in support of Bidder's design infrastructure and erection facilities and capacity and procedures including quality control related to the work, are enclosed at _[.....]_____ herewith.

Dated the *[Insert date of the month]* day of *[Insert month, year]*
at *[Insert place]*.

Signature {(of Bidder's authorized Bid Signatory (ies))[#] {In full and initials}}:

Full name: {insert full name of authorized Bid Signatory }

Title: {insert title/position of authorized Bid Signatory }

Name of Bidder (Sole Bidder's name or Consortium/ JV's name, if applicable):

Capacity: {insert the person's capacity to sign for the Bidder}

Address: {insert the authorized Bid Signatory's address}

Phone/fax: {insert the authorized Bid Signatory's phone and fax number, if applicable}

Email: {insert the authorized Bid Signatory's email address} _____

(Common Seal).....

Form 21**(Appendix to Technical Part of the Bid)****Format of Integrity Pact****INTEGRITY PACT****PRE-CONTRACT INTEGRITY PACT****GENERAL**

This pre-bid contract Agreement (herein after called the Integrity Pact) is made on _____ day of the month of _____ 2024, between, on one hand, the MAHARASHTRA STATE ELECTRICITY DISTRIBUTION COMPANY LIMITED

Acting through Shri....., Designation of the officer, (hereinafter called the MAHARASHTRA STATE ELECTRICITY DISTRIBUTION COMPANY LIMITED which expression shall mean and include, unless the context otherwise requires, his successors in the office and assigns) of the First Part and M/s. _____ represented by Shri _____ (hereinafter called the "BIDDER/SUPPLIER", which expression shall mean and include, unless the context otherwise requires, his successors and permitted assigns) of the Second Part.

WHEREAS the MAHARASHTRA STATE ELECTRICITY DISTRIBUTION COMPANY LIMITED propose to procure Plant and Installation Services as per the Scope of Work Mentioned in the RFB document (hereinafter called the "Facilities", against RFB No. Tender No. _____

[.....] for [Package Name/ Contract Title] which expression shall mean and include, unless the context otherwise requires, any additions & deletions in the said "Facilities") and the BIDDER/ Supplier is willing to offer/has offered the said "Facilities".

WHEREAS the BIDDER/ Supplier is a Private Company/Public Company/LLP/ Government Undertaking/ Partnership/Proprietorship, constituted in accordance with the relevant law in the matter and the MAHARASHTRA STATE ELECTRICITY DISTRIBUTION COMPANY LIMITED is a Ministry /Department of the Government of MAHARASHTRA [State Name] /SPSU performing its function on behalf of the Governor of MAHARASHTRA

NOW, THEREFORE,

To avoid all forms of corruption by following a system that is fair, transparent and free from any influence/prejudiced dealings prior to, during and subsequent to the currency of the Contract to be entered into with a view to:-

Enabling the MAHARASHTRA STATE ELECTRICITY DISTRIBUTION COMPANY LIMITED to obtain the desired "Facilities" at a competitive price in conformity with the defined specifications by avoiding the high cost and the distortionary impact of corruption on public procurement, and Enabling BIDDER/ SUPPLIER to abstain from bribing or indulging in any corrupt practice in order to secure the contract by providing assurance to them that

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their competitors will also abstain from bribing and other practices and the MAHARASHTRA STATE ELECTRICITY DISTRIBUTION COMPANY LIMITED will commit to prevent corruption, in any form, by its official by following transparent procedures.

The parties hereto hereby agree to enter into this Integrity Pact and agree as follows:
Commitments of the MAHARASHTRA STATE ELECTRICITY DISTRIBUTION COMPANY LIMITED

1.

The [MAHARASHTRA STATE ELECTRICITY DISTRIBUTION COMPANY LIMITED

- 1.1. undertakes that no official of the MAHARASHTRA STATE ELECTRICITY DISTRIBUTION COMPANY LIMITED connected directly or indirectly with the contract, will demand, take a promise for or accept, directly or through intermediaries, any bribe, consideration, gift, reward, favor or any material or immaterial benefit or any other advantage from the BIDDER/SUPPLIER, either for themselves or for any person, organization or third party related to the contract in exchange for an advantage in the bidding process, bid evaluation, contracting or implementation process related to the contract.
- 1.2. The Maharashtra State Distribution Company Limited will, during the pre-contract stage, treat all BIDDER/SUPPLIER alike, and will provide to all BIDDER/SUPPLIER the same information and will not provide any such information to any particular BIDDER/SUPPLIER which could afford an advantage to that particular BIDDER/SUPPLIER in comparison to the other BIDDER(S)/SUPPLIER(S).
- 1.3. All the officials of the Maharashtra State Distribution Company Limited will report the appropriate Government office any attempted or completed breaches of the above commitments as well as any substantial suspicion of such a breach.
2. In case any such preceding misconduct on the part of such official(s) is reported by the BIDDER/SUPPLIER to the Maharashtra State Distribution Company Limited with the full and verifiable facts and the same is prima facie found to be correct by the Maharashtra State Distribution Company Limited, necessary disciplinary proceedings, or any other action as deemed fit, including criminal proceedings may be initiated by the Maharashtra State Distribution Company Limited and such a person shall be debarred from further dealings related to the contract process. In such a case, while an enquiry is being conducted by the Maharashtra State Distribution Company Limited, the proceedings under the contract would not be stalled.

Commitments of BIDDERS/SUPPLIERS

3. The BIDDER/SUPPLIER commits itself to take all measures necessary to prevent corrupt practices, unfair means and illegal activities during any stage of its bid or during any pre-

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contract or post-contract stage in order to secure the contract or in furtherance to secure it and in particular commit itself to the following:-

- 3.1. The will not offer, directly or through intermediaries, any bribe, gift, consideration, reward, favour, any material or immaterial benefit or other advantage, commission, fees, brokerage or inducement to any official of the Maharashtra State Distribution Company Limited, connected directly or indirectly with the bidding process, or to any person, organization or third party related to the contract in exchange for any advantage in the bidding, evaluation, contracting and implementation of the contract.
- 3.2. The BIDDER/SUPPLIER further undertakes that it has not given, offered or promised to give, directly or indirectly any bribe, gift, consideration, reward, favour, any material or immaterial benefit or other advantage, commission, fees, brokerage, or inducement to any official of the Maharashtra State Distribution Company Limited or otherwise in procuring the Contract of forbearing to do or having done any act in relation to the obtaining or execution of the contract or any other contract with the Government for showing or forbearing to show favour or disfavour to any person in relation to the contract or any other contract with the Government.
- 3.3. The BIDDER/SUPPLIER shall disclose the name and address of agents and representatives and Indian BIDDERS shall disclose their foreign principals or associates.
- 3.4. The BIDDER/SUPPLIER shall disclose the payments to be made by them to agents/brokers or any other intermediary, in connection with this bid/contract.
- 3.5. The BIDDER/SUPPLIER further confirms and declares to the Maharashtra State Distribution Company Limited that he BIDDER/SUPPLIER is the original manufacture/Integrator/authorized government sponsored export entity of the defence stores and has not engage any individual or firm or company whether Indian or foreign to intercede, facilitate or in any way to recommend to the Maharashtra State Distribution Company Limited or any of its functionaries, whether officially or unofficially to the award of the contract to the BIDDER/SUPPLIER, nor has any amount been paid. Promised or intended to be paid to any such individual, firm or company in respect of any such intercession, facilitation or recommendation.
- 3.6. The BIDDER/SUPPLIER, either while presenting the bid or during pre-contract negotiations or before signing the contract, shall disclose any payment he has made, is committed to or intends to make to officials of the Maharashtra State Distribution Company Limited or their family members, agents, brokers or any other intermediaries in connection with the contract and the details of services agreed upon for such payments.
- 3.7. The BIDDER/SUPPLIER will not collude with other parties interested in the contract to impair the transparency, fairness and progress of the bidding process, bid evaluation, contracting and implementation of the contract.
- 3.8. The BIDDER/SUPPLIER will not accept any advantage in exchange for any corrupt practice, unfair means and illegal activities.
- 3.9. The BIDDER/SUPPLIER shall not use improperly, for purpose of competition or personal gain, or pass on to others, any information provided by the Maharashtra State

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Distribution Company Limited as part of the business relationship, regarding plans, technical proposal and business details, including information contained in any electronic data carrier. The BIDDER/SUPPLIER also undertakes to exercise due and adequate care lest any such information is divulged.

- 3.10. The BIDDER/SUPPLIER commits to refrain from giving any complaint directly or through any other manner without supporting it with full and verifiable facts.
- 3.11. The BIDDER/SUPPLIER shall not instigate or cause to instigate any third person to commit any of the actions mentioned above.
- 3.12. If the BIDDER/SUPPLIER or any employee of the BIDDER/SUPPLIER or any person acting on behalf of the BIDDER/SUPPLIER, either directly or indirectly, is a relative of any of the officers of the Maharashtra State Distribution Company Limited, or alternatively, if any relatives of an officer of the Maharashtra State Distribution Company Limited had financial interest/stake in the BIDDER's/SUPPLIER's firm, the same shall be disclosed by the BIDDER/SUPPLIER at the time of filling of tender.

The term 'relative' for this purpose would be as defined in Section 6 of the Companies Act 1956.

- 3.13. The BIDDER/SUPPLIER shall not lend to or borrow any money from or enter into any monetary dealings or transactions, directly or indirectly, with any employee of the Maharashtra State Distribution Company Limited.

4. Previous Transgression

- 4.1. The BIDDER/SUPPLIER declares that no previous transgression occurred in the last three years immediately before signing of this Integrity Pact with any other company in any country in respect of any corrupt practices envisaged hereunder or with any Public Sector Enterprise in India or any Government Department in India that could justify BIDDER's/SUPPLIER's exclusion from the tender process.
- 4.2. The BIDDER/SUPPLIER agrees that if it makes incorrect statement on this subject, BIDDER can be disqualified from the tender process or the contract, if already awarded, can be terminated for such reason.

5. Earnest Money (Security Deposit) (If applicable as per ITB Clause 18)

- 5.1. While submitting commercial bid, the BIDDER/SUPPLIER deposit an amount _____ (as specified in TENDER) as Earnest Money/Security, Deposit, with the Maharashtra State Distribution Company Limited through any of the following instruments:
 - 5.1.1. Bank Draft or a Pay Order in favour of Maharashtra State Distribution Company Limited' payable at [....]
 - 5.1.2. A confirmed guarantee by an Indian Nationalized Bank, promising payment of the guaranteed sum to the Maharashtra State Distribution Company Limited on demand within three working days without any demur whatsoever and without seeking any reasons whatsoever. The demand for payment by the Maharashtra

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State Distribution Company Limited shall be treated as conclusive proof of payment.

- 5.1.3. Any other mode or through any other instrument (to be specified in the TENDER.
- 5.2. The Earnest Money/ Security Deposit shall be valid as per terms of TENDER.
- 5.3. In the case of successful BIDDER/SUPPLIER, a clause would also be incorporated in the Article pertaining to Performance Bond in the procurement Contract that the provisions of Sanctions for violation shall be applicable for forfeiture of Performance Bond in case of a decision by the Maharashtra State Distribution Company Limited to forfeit the same without assigning any reason for imposing sanction for violation of this Pact.
- 5.4. No interest shall be payable by the Maharashtra State Distribution Company Limited to the BIDDER/SUPPLIER on Earnest Money/Security Deposit for the period of its currency.

6. Sanctions for Violations

- 6.1. Any breach of the aforesaid provisions by the BIDDER/SUPPLIER or any one employed by it or acting on its behalf (whether with or without the knowledge of the BIDDER/SUPPLIER/SUPPLIER) shall entitle the Maharashtra State Distribution Company Limited to take all or any one of the following actions, wherever required:-
 - 6.1.1. To immediately call off the pre contract negotiations without assigning any reason or giving any compensation to the BIDDER/SUPPLIER. However, the proceedings with the other BIDDER(s) would continue.
 - 6.1.2. The Earnest Money Deposit (in pre-contract stage) and/or Security Deposit/Performance Bond (after the contract is signed) shall stand forfeited either fully or partially, as decided by the Maharashtra State Distribution Company Limited and the Maharashtra State Distribution Company Limited shall not be required to assign any reason therefore.
 - 6.1.3. To immediately cancel the contract, if already signed, without giving any compensation to the BIDDER.
 - 6.1.4. To recover all sums already paid by the Maharashtra State Distribution Company Limited and in case of the Indian BIDDER/SUPPLIER with interest thereon at 2% higher than the prevailing Prime Lending Rate of State of India, while in case of a BIDDER/SUPPLIER from a country other than India with Interest thereon at 2% higher than the LIBOR. If any outstanding payment is due to the BIDDER from the Maharashtra State Distribution Company Limited in connection with any other contract for any other stores, such outstanding payment could also be utilized to recover the aforesaid sum and interest
 - 6.1.5. To en-cash the advance bank guarantee and performance bond/warranty bond, if furnished by the BIDDER/SUPPLIER, in order to recover the payments, already made by the Maharashtra State Distribution Company Limited, along with interest.
 - 6.1.6. To cancel all or any other contracts with the BIDDER/SUPPLIER. The BIDDER shall be liable to pay compensation for any loss or damage to the Maharashtra

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State Distribution Company Limited resulting from such cancellation/rescission and the Maharashtra State Distribution Company Limited /PRINCIPAL shall be entitled to deduct the amount so payable from the money(s) due to the BIDDER/SUPPLIER.

- 6.1.7. To debar the BIDDER/SUPPLIER from participating in future bidding processes of the Government of India Maharashtra State Distribution Company Limited PRINCIPAL for a minimum period of five years, which may be further extended at the discretion of the Maharashtra State Distribution Company Limited
- 6.1.8. To recover all sums paid in violation of this Pact by BIDDER/SUPPLIER (s) to any middlemen or agent or broken with a view to securing the contract.
- 6.1.9. In cases where irrevocable Letters of Credit have been received in respect of any contract signed by the Maharashtra State Distribution Company Limited with the BIDDER/SUPPLIER, the same shall not be opened.
- 6.1.10. Forfeiture of performance Bond in case of a decision by the Maharashtra State Distribution Company Limited to forfeit the same without assigning any reason for imposing for sanction for violation of this pact.
- 6.2. The Maharashtra State Distribution Company Limited will be entitled to take all or any of the actions mentioned at para 6.1 (i) to (x) of this Pact also on the Commission by the BIDDER/SUPPLIER or any one employed by it or acting on its behalf (whether with or without the knowledge of the BIDDER/SUPPLIER), of an offence as defined in Chapter IX of the Indian Penal code, 1860 or Prevention of Corruption Act, 1988 or any other statute enacted for prevention of corruption.
- 6.3. The decision of the Maharashtra State Distribution Company Limited to the effect that a breach of the provisions of this pact has been committed by the BIDDER/SUPPLIER shall be final and conclusive on the BIDDER/SUPPLIER. However, the BIDDER/SUPPLIER can approach the Independent Monitor(s) appointed for the purposes of this Pact.
7. deleted
 - 7.1. deleted
8. Independent Monitors
 - 8.1. The Maharashtra State Distribution Company Limited has appointed Independent Monitors (hereinafter referred to as Monitors) for this Pact in consultation with the Central Vigilance Commission (Names and Addresses of the Monitors shall be published subsequently by Maharashtra State Distribution Company Limited
 - 8.2. The task of the Monitors shall be to review independently and objectively, whether and to what extent the parties comply with the obligations under this Pact.
 - 8.3. The Monitors shall not be subject to instructions by the representatives of the parties and perform their functions neutrally and independently.
 - 8.4. Both the parties accept that the Monitors have the right to access all the documents relating to the project/procurement, including minutes of meetings.

- 8.5. As soon as the Monitor notices, or has reason to believe, a violation of this Pact, he will so inform the Authority designated by the Maharashtra State Distribution Company Limited
- 8.6. The BIDDER/SUPPLIER(s) accepts that the Monitors has the right to access without restriction to all project documentation of the Maharashtra State Distribution Company Limited including that provided by the BIDDER/SUPPLIER. The BIDDER/SUPPLIER will also grant the Monitor, upon his request and demonstration of a valid interest, unrestricted and unconditional access to his project documentation. The same is applicable to Subcontractors. The Monitor shall be under contractual obligation to treat the information and documents of the BIDDER/SUPPLIER/Subcontractors(s) with confidentiality.
- 8.7. The BUYER will provide to the Monitors sufficient information about all meetings among the parties related to the Project provided such meetings could have an impact on the contractual relations between the parties. The parties will offer to the monitor the option to participate in such meetings.
- 8.8. The Monitor will submit a written report to the designated Authority of Maharashtra State Distribution Company Limited] /Secretary in the Department/ within 8 to 10 weeks from the date of reference or intimation to him by the Maharashtra State Distribution Company Limited BIDDER/SUPPLIER and, should the occasion arise, submit proposals for correcting problematic situations.

9. Facilitation of Investigation

In case of any allegation of violation of any provisions of this Pact or payment of commission, the Maharashtra State Distribution Company Limited or its agencies shall be entitled to examine all the documents including the Books of Accounts of the BIDDER/SUPPLIER and the BIDDER/SUPPLIER shall provide necessary information and documents in English and shall extend all possible help for the purpose of such examination.

10. Law and Place of Jurisdiction

This Pact is subject to Indian Law. The place of performance and jurisdiction is the seat of the Maharashtra State Distribution Company Limited

11. Other Legal Actions

The actions stipulated in this Integrity Pact are without prejudice to any other legal action that may follow in accordance with the provisions of the any extent law in force relating to any civil or criminal proceedings.

12. Validity

- 12.1. The validity of this Integrity Pact shall be from the date of its signing and extend up to 5 years or the complete execution of the contract to the satisfaction of both the Maharashtra State Distribution Company Limited and the BIDDER/SUPPLIER, including warranty period, whichever is later. In case BIDDER/SUPPLIER is

unsuccessful, this Integrity Pact shall expire after six months from the date of the signing of the contract.

12.2. Should one or several provisions of this Pact turn out to be invalid; the remainder of this Pact shall remain valid. In this case, the parties will strive to come to an agreement to their original intentions.

13. The parties hereby sign this Integrity Pact at _____ on _____

Signed on(*Insert the Date*)

Signature (of Bidder's authorized Bid Signatory)[#] {In full and initials}:

Full name: {insert full name of authorized Bid Signatory}

Title: {insert title/position of authorized Bid Signatory}

Name of Bidder (Sole Bidder's name or Consortium/ JV's name, if applicable):

Capacity: {insert the person's capacity to sign for the Bidder}

Address: {insert the authorized Bid Signatory's address}

Phone/fax: {insert the authorized Bid Signatory's phone and fax number, if applicable}

Email: {insert the authorized Bid Signatory's email address} _____

[#]{*For a joint venture, either all members shall sign or only the authorized signatory as per ITB 19.4, in which case the power of attorney to sign on behalf of all members shall be attached*}

Witness

1. _____

2. _____

[UTILITY NAME]L

Name of the Officer

Designation

Deptt/MINISTRY/PSU

Witness

1. _____

2. _____

* Provision of these clauses would need to be amended/deleted in line with the policy of the [UTILITY NAME] in regard to involvement of Indian agents of foreign supplies.

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Section - 5 : Bidding Forms - Financial Part of the Bid

INSTRUCTIONS TO BIDDERS: The bidder must fill and submit the price schedules as specified in BDS.

PART 2

EMPLOYER'S REQUIREMENTS

Section - 6 : Employer's Requirements

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1. Overview of the Scope of Works

The work is to be executed on turnkey basis, the scope of which includes survey, network design, supply, manufacturer's quality assurance, testing (where specified/ required), transportation, storage, erection, including all civil/ structural works, site testing, commissioning of all items & materials including all associated activities though not exclusively specified herein and are required for the completion and satisfactory performance of the entire works as intended.

The scope of works also include General Technical Instructions enclosed at Annexure-B.

This specification intends to cover but not limited to the following activities, services and works:

- Providing engineering data, and drawing for review, approval and records.
- Supply, testing, packing, transportation and insurance from the manufacturer's work to the site.
- Receipt, storage, insurance, preservation and conservation of equipments at the site.
- Fabrication, pre-assembly (if any), erection, testing and putting into satisfactory operation of all the equipments/ materials including successful commissioning.
- In addition to the requirements indicated in this section, all the requirements as stated in Technical specifications shall also be considered as a part of this specification as if completely bound herewith.
- Providing all materials, equipments and services specified or otherwise, including survey, which are required to fulfill the intent of ensuring operability, maintainability and the reliability of the complete works covered under this specification.
- During warranty period the bidder is required to provide all the services and activities mentioned in the contract.
- Preparation of project completion report/ closure proposal along with a report clearly indicating completion of any outstanding/ remedial work that needs to be carried out.
- Handing over the works to the employer for taking into commercial services.

It is not the intent to specify all aspects of design and construction of equipments mentioned herein. The systems, subsystems and equipments shall conform in all respect to high standards of engineering, design and workmanship and shall be capable of performing in continuous commercial operation. Accordingly, scope of works under this contract is as under:

- 1.1 Execution of all other works as per tender document. All Steel structure except STPs shall be hot dip galvanized as per relevant Indian Standard.
- 1.2 All the raw materials such as steel, zinc for galvanizing, reinforcement steel and cement for foundation, coke for earthing, bolts, nuts & washers, danger plates, phase plate, number plate etc. required for substations & its structures shall be included in the scope of supply. Bidders shall clearly indicate in their offer, the sources from where they propose to procure the key raw materials and the components.

- 1.3 The Technical Specifications of Equipments are as per MSEDCL Technical Specifications published on MSEDCL website .
- 1.4 A set of drawings are published on the MSEDCL website. These are tender drawings and are to be approved by the Employer. These drawings are indicative in nature and therefore, must be referred while preparing drawings for approval.
- 1.5 All the new assets created under System Strengthening to be properly GIS tagged with the help of mobile App provided by the Employer.
- 1.6 The engraving of word “Developed under MPDEP (ADB) Scheme” in materials viz., Poles, Transformers (All types), Cables, Energy Meter etc is mandatory requirement. The Employer shall ensure strict compliance of this requirement. Also, while processing payments to the Contractor, suitable documentary evidence / photographs must be asked by the Employer in support of the compliance.
- 1.7 Major infrastructures like new primary substation, new Distribution Transformer, new lines etc developed under System Strengthening needs to be clearly denoted by a signboard that should represent
 - 1.7.1. About the Work
 - 1.7.2. Date of Commissioning
 - 1.7.3. Estimated cost
 - 1.7.4. Scheme
 - 1.7.5. Employer's details
 - 1.7.6. For Transformers, rating in kVA

The signboard shall comply to the technical specifications attached and shall reflect as a separate billable item in the price schedule attached . Colour coding of the DTs is essential.

- 1.8 The Manufacturing Quality Plan (MQP) shall be finalized by the Employer in consultation with the contractor/manufacturer in line with the Technical Specifications and as per Employer's practices

Supply of Plant and Services under this tender covers all interventions required for satisfactory operations of the facilities unless specifically excluded. Scope includes Design, Supply, survey, installation, erection testing and commissioning, on turnkey basis. The types of works envisaged in the System Strengthening scheme are:

A. Modernization and system strengthening:

- New 33/11 kV, 22/11 kV, 33/22 kV, 66/33 kV Primary Substation (as applicable)
- New 33 kV Lines
- New 22 KV Lines
- New 11 kV Lines
- Augmentation of Power Sub-station and 33kV, 22kV, 11kV line
- Renovation
- DTR Substations
- LT Line
- Under-ground cabling works
- Installation of Capacitor Banks

All items to be supplied and erected shall be strictly as per the specifications given in the Bid and should comply the relevant standards and any amendments thereof.

Any deviation taken by the bidder and not specifically / clearly brought out in the price schedule will not be considered as a valid deviation.

In addition to the works mentioned above the bidder is required to take care of the activities listed below:-

1.9 Quantity Variation: Activity-wise Scope in the tender may vary according to site conditions, urgency of work and the Competent Authority / Tender issuing authority has reserved the right to Add or Reduce the Quantity of any of the activities fully or partially.

1.10 Credit of Material: Material used for providing temporary / provisional supply or material removed during the execution of replacement / upgradation activities shall be credited by the Contractor to MSSEDCL Store in good condition & well within time by following prevailing procedure.

2. Survey

The Contractor shall carry out, and be responsible for, final design of the works, including any site surveys, subsoil investigations and all other things necessary for proper planning design and execution. The initial site surveys will be carried out for tentative freezing of the material requirement and the work content finalization, within one month of commencement of project, and this will be a joint survey, along with the employer. The same shall be reviewed progressively on quarterly basis for freezing of the material requirement and work content. Design shall be prepared by qualified designers who are engineers and experienced in design of transmission and distribution systems. Employer shall provide all options proposed for loss reductions to the contractor. Contractor, while surveying the execution of work, shall keep this requirement in view and suggest best loss reduction options in descending order. Means, maximum loss reduction option shall be proposed on priority. Also, while executing the works, same

priority of works must be followed.. The changes in design should be approved by Engineer-in-charge of Employer.

- Based on the finalized network design, the Bidder shall carry out the field survey for deciding location of poles, distribution transformers etc. Besides, field survey will also cover the following:
 - a. Proposed route of 33 KV, 11KV and LT Line.
 - b. Proposed location of 33/11 KV sub-stations along with number, capacity of power transformers and number of 11 KV feeders to be taken out.
 - c. Locations of new distribution transformers.
- Feeder wise Bill of Quantity (BOQ), for each proposed work will be prepared by the Bidder and submitted to the Employer's Engineer-in-charge for his approval before commencement of actual work.
- All architectural and civil designs such as control room, foundation for equipment, transformer, isolators, VCB shall be approved by the Employer's Engineer-in-charge.
- Within substation yard, all clamps and connectors used for bus-bar will be crimped. All the clamps and connectors connecting the equipments shall be of bolted type.

3. Project Management System

3.1. General

The Contractor shall assign a project manager with the authority to make commitments and decisions that are binding on the Contractor. Employer will designate an officer incharge to coordinate all employer project related activities. All communications between employer and the Contractor shall be coordinated through the project manager and officer incharge of Employer. The project managers shall also be assisting employer in communicating project related information to other stake holders.

Bidder shall submit the manpower deployment plan along with the bids, describing the key roles of each person.

The role and responsibilities of contractor shall be as follows:

- a) To prepare, maintain and update project detailed Work Execution Plan for successful implementation of project like approval of GTP, approval of sub-contractor, approval of drawings, supply of materials, mobilization of men, material and equipment etc. at site for successful completion of works, Compile and up-load physical as well as financial progresses, compile the progress of works at Employer level and to assist in forwarding it to all stake holders.

- b) To actively participate with employer in resolving all issues relating to project implementation including ROW, Forest Clearances and Railway Crossings.
- c) To actively participate in monitoring, reviewing and analysing the physical, financial and quality assurances works' progress of System Strengthening works and also to take suitable measures on compliance of observations being raised during monitoring/review meetings with employer.
- d) To upload and up-date project wise physical progress in System Strengthening web portal. Physical as well as financial progresses shall be uploaded in standard Bill of Material format of the contract. Also, to submit invoices as per released System Strengthening guidelines to Employer for release of payments/funds.
- e) To oversee the progress and compliance of the Quality Assurance Mechanism as per System Strengthening guidelines.

3.2. Project Schedule

As per the schedule the bidder shall submit a preliminary implementation plan along with the bid. The detailed project implementation schedule shall be submitted by the contractor after the award of contract for employer's approval, which shall include at least the following activities:

- (a) Surveying of site.
- (b) Documents submission and approval schedule
- (c) Pre-Dispatch Inspection schedule
- (d) Dispatch Schedule
- (e) Installation & commissioning schedule
- (f) Training schedule, if any.

The project schedule shall include the estimated period for completion of project and its linkage with other activities etc. It is expected that the contractor should share updated project schedule based on the actual progress done at site, priorities of the employer, availability of material etc once per quarter along with the Progress report.

3.3. Progress Report

A progress report shall be prepared by the Contractor each month against the activities listed in the project schedule. The report shall be made available to employer on a monthly basis, e.g., the 10th of each month. The progress report shall include all the completed, ongoing and scheduled activities.

3.4. Transmittals

Every document, letter, progress report, change order, and any other written transmissions exchanged between the Contractor and employer shall be assigned a unique transmittal number. The Contractor shall maintain a correspondence index and assign transmittal numbers consecutively for all Contractor documents. Employer will maintain a similar correspondence numbering scheme identifying documents and correspondence that employer initiates.

4. Quality Assurance and Evaluation Mechanism

The Quality Assurance (QA) will be carried out by Employer. The Employer may engage an Authorised representative of employer responsible & accountable for assuring quality in System Strengthening works. Key activities would include:

- Formulation of a detailed comprehensive Quality Assurance Mechanism (QAM) plan/Guaranteed Technical Particulars as the case may be in the State for the works to be carried out under System Strengthening scheme with an objective to create quality infrastructure works. The QAM and Inspection Plan shall be integral part of the contract agreement with turnkey contractor or equipment supplier and erection agency as the case may be in case of turnkey/ or departmental execution of works.
- Ensuring that the quality of materials/equipment supplied at site and execution of works carried out at field under System Strengthening scheme is in accordance to Manufacturing Quality Plan (MQP)/Guaranteed Technical Particulars (GTP) and Field Quality Plan (FQP)/Approved Drawings/Data Sheets respectively.

4.1. Quality checks to be ensured by Turnkey Contractor:

Turnkey Contractor shall strictly ensure QAM checks during the day to day course of project execution, which are as follows:

- a. Pre-dispatch inspections of all materials viz. as per MQP/GTP, Approved Drawings, Technical Specifications, Datasheet, GTP, applicable national & international standards as per GCC Clause 23.
- b. 100% verification of all 66/11 or 33/11 kV sub stations (New & Augmented) for quality of material as per MQP/GTP, Approved Drawings, Technical Specifications, Datasheet and erection works in the field as per FQP/approved drawings.
- c. 100% verification of feeders created under the scheme.
- d. 100% verification of materials utilised under the scheme.
- e. 100% verification of works done in Loss Reduction and Modernization & System Strengthening.

4.1.1. Vendor approval: All the materials procured for System Strengthening works shall be purchased from the authorised vendors approved by the Quality Assurance Department of Employer. Approved vendors list is uploaded periodically on the Employer web portal.

New vendors/suppliers may be approved by Employer, provided capability of manufacturer(s) is assessed suitably by visiting the factory premises and checking the testing facility available before accepting it as an approved vendor. If required, State Electricity Board/Power Department/Distribution Companies may adopt vendors already approved by CPSEs.

4.1.2. FQP for Civil works: Employer shall prepare a separate FQP/field execution drawings which shall be approved by their competent authority which shall be uploaded at web portal. The turnkey shall adhere to this FQP/drawings while carrying out physical works. Contractor

4.1.3. FQP for testing & commissioning: Employer shall prepare a comprehensive Pre-commissioning test Check-list for testing & commissioning of 33/11kV or 66/11kV substation, Distribution transformer Substation etc. The electrical system shall be energized only after performing all tests as described in the pre-commissioning test checklist. and inspection from the electrical inspector of the state (or as the practice may be). Proper records in this regard, including tests on earth resistance, insulation resistance of 11 kV line & Distribution Transformer etc. shall be maintained, jointly signed by Employer and turnkey representatives.

4.1.4. Quality Assurance Mechanism (QAM) to be followed by the Contractor is as below:

- a. The Contractor shall be responsible and accountable for assuring quality in the scheme works. Accordingly, the Contractor shall formulate a comprehensive Quality Assurance mechanism (QAM) and Inspection Plan with an objective to build quality infrastructure under the project, which should be approved by the Employer. Alternately, the Employer may also provide its QAM which needs to be complied by the Contractor. The QAM and Inspection Plan shall be an integral Part of the contract agreement with turnkey Contractor or equipment supplier/vendor and erection agency as the case may be in case of partial turnkey and departmental execution of works. Documentation with regard to Quality Assurance and Inspection Plan shall be maintained by the Contractor and kept in proper order for scrutiny during the course of project execution and for future reference. The Contractor has to ensure that the quality of materials/equipment's supplied at site and execution of works carried out at field is in accordance to the Manufacturing Quality Plan (MQP)/Guaranteed Technical Particulars (GTP) and Field Quality Plan (FQP)/Approved Drawings/Data Sheets respectively.
- b. Some key indicative measures for effective implementation of the QAM by the Contractor are given below. However, these are for reference and need to be followed as per relevant provisions of the contract.
 - Supply:

- Verification of qualifications of the subcontractor / manufacturer for supply of plant / equipment and materials. Factory inspections may be conducted if required.
- Verification of material data, specifications, drawings and samples submitted by the subcontractor / manufacturer including GTPs.
- Verification of type test reports including qualifications of the test laboratory, completeness and acceptance of the type test reports.
- Witnessing acceptance tests carried out by the subcontractor/ manufacturer.
- Carrying out pre-dispatch inspections as per relevant guidelines of this tender/ contract.
- Inspection of storage facilities of the subcontractor/ manufacturer.
- Works:
 - Carry out field inspections on sample basis during implementation to verify works are carried out in compliance to technical specifications and acceptable quality of workmanship.
 - Issue Site Observation Reports (SOR) and follow-up with the subcontractor/ manufacturer for implementation of any remedial actions.
 - Upon completion, carry out joint inspections together with the Employer's staff and for final measurements and quality inspections.
 - Follow-up any on technical issues for corrective action during defects liability period with the subcontractor/ manufacturer.
- c. It should be noted that no functional guarantees are applicable for equipment installed as a part of this contract hence Guarantee Tests are not applicable.
- d. The Employer may identify any third party agency including PMA/ TPQMA etc. who would be responsible to monitor the QAM measures including verifications and inspections mentioned above. The project manager may also engage third party inspectors for this purpose in addition if required.
- e. The Employer or its appointed third party shall design systems and procedures to implement QAM system including formats for submittals by the Contractor in line with the above requirements and provisions of the Contract.
- f. The Contractor shall cooperate with and follow these QAM systems and procedures to ensure proper implementation of an effective quality assurance and evaluation mechanism.

4.1.5. Pre-commissioning test record: All pre-commissioning test check list shall be documented properly and signed by the quality engineer of the turnkey Contractor & countersigned by Employer's representative and shall be kept for future reference. These documents shall be maintained by Employers in proper order and shall be made available at site for verification by

Quality Monitors during inspection and finally be handed over to user department O&M department) at the time of handing over of energized assets.

4.1.6. Roles and responsibility of Contractor in ensuring Quality of Plant and Facilities:

- 1) Turnkey contractor shall be primarily be responsible for supply of quality materials. Hence, turnkey contractor shall take all necessary actions including following:
 - a. To assess the capabilities and capacity of manufacturer to whom they intend to appoint as sub-vendor,
 - b. To keep strict control over manufacturing of materials by controlling procurements of right raw materials, periodical stage inspections, to ensure process control and to get the materials invariably inspected in manufacturing stage as well as after manufacturing but before dispatch at the works of manufacturer to ensure quality of materials/equipment.
 - c. To ensure stage inspection and final dispatch inspection, turnkey contractor should deploy his/her quality assurance team to inspect the materials with Employer/third party inspector as well as independently as per requirements.
- 2) Receipt inspection: On receipt of materials at site, it would be the prime responsibility of turnkey contractor to verify materials physically in accordance to agreed technical specifications. Physical parameters like dimensions (length, width, height, area of conductor), weight per unit, Insulation Value, length of cable/conductor in sample drum(s), clear embossing on cables through sequential marking depicting name of manufacturer, size of cable and length in meter. Once the Contractor is satisfied, materials must be offered for joint inspection to Employer.
- 3) Earlier, on receipt of materials at site, dispatch documents shall be verified jointly by Employer, Employer's appointed Third Party, Turnkey Contractor and materials supplier (if representative is full time available at site. During inspection, quantities of items, sealing on the materials, serial numbers of the items, sequential embossing (proper visible/legible without any additional efforts) and name plates on the materials shall be checked. Dispatch challans shall be verified for details of consigner and consignee, materials descriptions, quantities transported, pre-dispatch clearance certificate/waiver of inspection. In case of high value equipment, capacity of equipment in terms of current carrying capacity, operating voltage and KVA ratings should be recorded.
- 4) Clearance for installation: Once, materials on receipt are accepted by turnkey contractor as well as Employer representative, they will be eligible for erection, testing and commissioning.
- 5) Sampling from field: Any material, including materials listed below, may be picked from site for testing at test laboratory chosen by inspecting official. 1. Distribution Transformer, 2. Overhead Conductor, 3. Energy Meter, 4. Pole, 5. Insulators, 6. Cables, 7. Circuit Breaker.

- a. Inspecting official will have right to pick any of the supplied equipment whether it is lying in site stores, is under erection, is under local transportation from site stores to erection location or is already commissioned. The equipment shall be sealed jointly in presence of representatives of Employer, Employer's appointed Third Party, Contractor, and Supplier/manufacturer (if his representative is available at site). Employer at its discretion may invite manufacturer's representative to participate in sealing of materials.
- b. Sealed equipment, on cost of turnkey contractor shall be sent to test laboratory for verification of routine/type test results. At the time of sealing, details of equipment available at site shall be recorded like cable/conductor drum number, power/distribution transformer unique number, status of sequential legible embossing on cables, name of manufacturer etc.
- c. For testing of equipment, Employer/Nodal agency shall empanel test laboratories located in or nearer to the state wherefrom sample is picked up.
- d. Such picked up materials at a random shall be tested for all routine, acceptance and type tests feasible to conduct in the empaneled laboratory. The laboratory expenses including all other expenditures that shall incurred towards packing, transport, inspection, testing charges etc. are to be borne by the Employer. At least one sample from a lot may be subjected to inspection.
- e. In cases, where pre-dispatch and factory test results/NABL accredited lab test report are found mismatched with tests results on the sample picked from field; actions shall be taken against willful defaulted manufacturer and turnkey contractor both.
- f. Willful defaulter shall be those manufacturers and turnkey contractor whose material is found to be manufactured using inferior quality raw materials, second hand core materials, under-size/under-weight of cable/conductor in various parts of cable/conductor drum, not conforming to transformer load losses as defined in agreed technical specifications/contract conditions, improper or no sequential legible embossing on cable etc.
- g. This mismatch shall generally be, but not limited to the deviations in results from guaranteed technical specifications of materials in terms of capacity (KVA capacity, current carrying capacity, heating capacity, tensile strength, mechanical strength etc., operational efficiencies (errors in measurements of power, power/load losses, power consumption etc., weights of key component materials (aluminum, copper, insulation materials, steel components etc.), sub-standard specifications of key component (measured specifications are in deviation from guaranteed specifications as per technical specifications of the project and inferior/illegible embossing/sequential marking on cables are found, following two actions shall be taken:
 - i. Sub-vendors/vendor registration of the manufacturer shall be discontinued in all the power utilities of the country for a period of 5 years including in power utility concerned where this act is found,

- ii. Entire lot of materials/equipment supplied by the defaulting manufacturer shall be rejected whether supplied materials/equipment is lying in site-stores, in transit, under erection, testing & commissioning or has already been commissioned. All costs related to removal of such rejected materials and reinstating fresh lot of materials shall be borne by turnkey contractor without any cost implication to power utility.
 - iii. Turnkey contractor shall be responsible for repetitive failures of materials in field testing in a turnkey-contract. In such situations, registration of turnkey contractor firm shall be discontinued in all the power utilities of the country for a period of 5 years including in power utility concerned where this act is found,
 - h. In cases, where field testing results are slightly mismatched with factory test results / pre-dispatch test reports/NABL accredited lab test report but are in permissible limits as per GTP/Data Sheet/Technical Specifications, no action shall be taken against the turnkey contractor/manufacturer.
 - i. In cases where turnkey contract is reluctant/not willing to support the Employer in selecting sample for testing by way of non-association in sampling, sampling and testing related activities of equipment, all actions related to sample selection, sealing and testing including dismantling, loading, unloading, transportation etc, will be taken by Employer on risk & cost of the turnkey contractor. The non-cooperative act on part of turnkey contractor shall be circulated amongst all power utilities in the country. In such situations, registration of turnkey contractor firm shall be discontinued in all the power utilities of the country for a period of 5 years including in power utility concerned where this act is found.
- 6) <Optional clause – to be deleted>
- ~~The Contractor should set up at least one testing laboratory under this contract for testing the materials received at site from its subcontractors/ vendors/ manufacturers etc. The testing laboratory thus setup, should have the facility for carrying out basic tests to ascertain the quality of the following equipments/ materials:~~
- ~~a. Cables/ Conductors~~
 - ~~b. Distribution Transformers~~
 - ~~c. Insulators~~
- ~~[100%] quantity of each lot shall be tested by the Contractor at its field test laboratory. Post completion of testing of the above items at its laboratory, the Contractor shall submit a report to the Employer, certifying the satisfactory testing results.~~
- 7) The Contractor should develop the quality assurance programme which shall generally cover the following:
- a. Organization structure for the management and implementation of the proposed quality assurance programme :

- b. Documentation control system;
 - c. Qualification data for bidder's key personnel;
 - d. Procedure for purchases of materials, parts, components and selection of sub-Contractor's services including vendor analysis, source inspection, incoming raw material inspection, verification of material purchases etc.
 - e. System for shop manufacturing and site erection controls including process controls and fabrication and assembly control;
 - f. Control of non-conforming items and system for corrective actions;
 - g. Inspection and test procedure both for manufacture and field activities.
 - h. Control of calibration and testing of measuring instruments and field activities;
 - i. System for indication and appraisal of inspection status;
 - j. System for quality audits;
 - k. System for authorizing release of manufactured product to the Employer.
 - l. System for maintenance of records;
 - m. System for handling storage and delivery; and
 - n. A manufacturing quality plan detailing out the specific quality control measures and procedures adopted for controlling the quality characteristics relevant to each item of equipment furnished and/or services rendered.
 - o. A Field quality Plan covering field activities
- 8) **Electrical Inspector inspection:** After successful completion of the work, the contractor shall take permission from State Electrical Inspectorate. In case of defects / in-complete works notified by Electrical Inspectorate, these shall be completed by the agency at no extra cost implication to Employer.

4.2. Material Inspection: For the purpose of inspection, materials have been segregated into two categories as mentioned below:

4.2.1 Category – A (Pre-Dispatch Inspection & Testing at NABL accredited Labs):

- a) This category shall include high ticket materials (Power Transformers, Distribution Transformers, Circuit Breakers, AB/XLPE Cables, Overhead Conductor (AAAC/ACSR), Insulator which involves more and important testing procedures and hence the inspection of these materials will be carried out in the factory before the dispatch of the material.
- b) In addition, Employer shall also ensure that for major materials as discussed above, samples from 1st lot and one other lot randomly selected by the Employer shall be directly sent to NABL accredited test labs for third party testing. It is also to be noted that material clearance of the lots under testing shall only be given post receipt of successful test results. Contractor shall also mandatorily depute its authorized official for pre – dispatch inspection at manufacturing facility alongwith the Employer officials. The inspection and testing report would be jointly signed by the

Employer and the Contractor. All the expenses related to testing would be beared by the Contractor.

- c) Apart from the above mentioned protocol any one power transformer shall be selected by Employee from the supply schedule from the vendor, which shall be jointly sealed and tested for short circuit testing on turnkey-Contractor's expenses.

4.2.2 Category – B (On-site inspection): This category includes the materials for which a factory inspection is not warranted and the material can be inspected upon arrival at the site before the installation. In case the Employer is apprehensive about the quality of the material supplied it reserves the right to send the selected lot to the NABL accredited testing lab for third party testing.

4.2.3 Employer also reserves the right to send any installed equipment / materials to the NABL accredited testing lab for testing. The Employer would have to reimburse the expenses related to transportation of material from site to testing lab and all testing expenses in this regard.

4.2.4 The material which has to be tested at laboratory shall be sealed in the presence of authorized official of Employer and Contractor.

4.2.5 If the materials tested at Laboratory fails then the entire lot would be rejected. Contractor shall bear the responsibility of sending back such failed materials from site. Any subsequent delay in contract performance due to failure of materials in the test laboratory would be on account of Contractor and no time extension would be provided by the Employer in this regard. Any LD levies in this regard would be borne by the Contractor.

4.2.6 Pre-dispatch Inspection (PDI) for Category-A

4.2.6.1. Pre-dispatch inspection shall be performed on the identified materials at manufacturer's work place for which Contractor shall be required to raise requisition giving at least 10-day time. Depending on requirement, inspection shall be witnessed by representatives of Employer, TPQMA, Contractor and/or representative of the Nodal Agency. TPQMA shall carry out pre-dispatch inspection of major materials randomly in a single lot containing minimum 10% materials at manufacturer works. The TPQMA shall perform pre – dispatch inspection test of below materials:

- 1) Power Transformer,
- 2) Circuit Breaker,
- 3) Insulators,
- 4) Cables / Conductors
- 5) Control & Relay Panel,
- 6) Distribution Transformer

4.2.6.2. Tests to be conducted during the Pre-dispatch Inspection: All the tests shall be carried out in accordance with the latest relevant IS published from time to time by BIS. An indicative list of IS specification and tests for some of the materials are given below:

Power Transformers

Standards Applicable:-	
IS:2026 (Part I to IV)	Power Transformer
IS:5	Colour for ready mixed paints
IS:325	Three Phase Induction Motors
IS:335	New insulating oil for transformers, switch gears
IS:1271	Classification of insulating materials for electrical machinery and apparatus in relation to their stability in services
IS:2071	Method of high voltage testing
IS:2099	High voltage porcelain bushings
IS:2147	Degree of protection
IS:2705	Current Transformers
IS:3202	Code of practice for climate proofing of electrical equipment
IS:3347	Dimensions for porcelain Transformer Bushings
IS:3637	Gas operated relays
IS:3639	Fittings and accessories for power Transformers
IS:5561	Electric Power Connectors
IS:6600/BS:CP "10:0	Guide for loading of oil immersed Transformers
IS:10028	Code of practice for selection, installation and maintenance of transformers, Part I. II and III
C.B.I.P. Publication	Manual on Transformers
Proposed Tests as per IS	
1	All standard routine tests in accordance with IS: 2026 with dielectric tests corresponding to latest amendments of IS: 2026 shall be carried out.
2	All auxiliary equipment shall be tested as per the relevant IS. Test certificates shall be submitted for bought out items.
3	High voltage withstand test shall be performed on auxiliary equipment and wiring after complete assembly.
4	Following additional routine tests shall also be carried out on each transformer: a) Magnetic Circuit Test on each core shall be tested for 1 minute at 2000 Volt DC b) Oil leakage test on transformer

Distribution Transformers

Standards Applicable:-

IS 2026: (Part 1 to 10) as applicable	General Specification of Transformer
IS 1180 (Part1) (Including Amendment 1, 2, 3 & 4)	Outdoor type oil immersed Distribution transformer upto including 2500 kVA & 33 kV
IS 3347 (Part 1)	Specification upto 1.1 kV Voltage Bushing (for Porcelain)
IS 3347 (Part 3)	Specification upto 17.5 kV Voltage Bushing (for Porcelain)
Type Test	A valid Type test report within 5 years of supply 3 star rated transformer as per BEE
IS -5484	Specification for Aluminum wire rods
IS 12444	Specification for Copper wire Rods
Proposed Tests as per IS	
1	Measurement of Winding resistance at all Tap Positions
2	Ratio of Tap Position Polarity & Phase Position
3	% Impedance at 75 degree Celsius at 50Hz
4	Load losses at 50% and at 100% Loading on Energy efficiency Level -5 as per IS 1980 (Part-1) 2014 and its amendment 1.2. 3 and 4
5	No Load Loss at 50Hz and No load current at rated voltage
6	Insulation Resistance
7	Induced Overvoltage Withstand
8	Separate source voltage Withstand
9	Magnetizing Current at Rated voltage and frequency an 112.5% of rated voltage
10	Temperature rise test
11	Pressure Test
12	Oil Leakage Test

Circuit Breaker

Standards Applicable:-	
IS: 13118	Specification for HV AC Circuit Breaker
IS: 14658	HV AC Circuit Breakers - Guide for Short-circuit and Switching Test Procedures for Metal-enclosed and Dead Tank Circuit Breakers
IS: 2099	Specification for HV porcelain bushings
IS: 5621	Specification for porcelain hollow insulator.
IS: 8603	Specification for Dimension for Porcelain Transformer Bushing for use in heavily polluted area.
IS: 3347	Specification for Dimension for Porcelain Transformer Bushing for use in normal and lightly polluted area.
IS: 2633	Specification for method for testing uniformity of coating On Zinc coated articles.
IS: 5561	Specification for Electrical Power Connectors
IS: 2147	Specification for Degree of Protection

Proposed Tests as per IS

1	Single Capacitor bank breaking test
2	Short time withstand and Peak Withstand Current Test
3	Wet power frequency withstand voltage test
4	Lightening Impulse voltage withstand test
5	Temperature rise test
6	Mechanical Endurance Test(M2 Class)
7	Degree of Protection test of Control Cubicle

XLPE Cables

Standards Applicable:-	
IS : 7098 (Part-I) : 1988	Specification for Crosslinked Polyethylene Insulated PVC sheathed Cables for working Voltage upto & including 1100 Volt
IS: 8130 : 1984	Specification for Conductors for insulated electric cables and flexible cords
IS:5831 : 1984	PVC insulation & sheath of electric cables
IS: 3975 : 1970	Specification for Low Carbon Galvanized steel wires, Formed Wires and tapes for armouring of Cables
IS:10810 : 1984	Methods of test for Cables.
IS:10418 : 1982	Cable Drums for Electric Cables.
IS : 694 : 2010	PVC unsheathed and Sheathed cables / Chords with rigid and flexible conductor for rated voltages upto and including 450/ 750 V
Proposed Tests as per IS	
1	Resistance Test on conductor
2	Test for thickness of insulation and sheath
3	Tensile strength and elongation at break test for insulation & outer sheath.
4	Hot set test for insulation.
5	Insulation resistance (Volume resistivity) Test
6	High voltage test

Aerial Bunched Conductor

Standards Applicable:-	
IS 14255 : 1995	Specification for Aerial Bunched Cables for working Voltage up to & including 1100 Volts.
IS : 8130 : 1984	Specification for Conductors for Insulated Electric Cables and flexible Cords.
IS : 398 (Part-IV) : 1994	Specification for Aluminum Conductors for overhead transmission purposes: Aluminum Alloy Stranded Conductors (Aluminum Magnesium –Silicon type).
IS:10418:1982	Specification for Drums for Electric Cables
Proposed Tests as per IS	
1	Breaking Load Test for messenger Conductor.

2	Elongation test for messenger conductor.
3	Conductor resistance test for both messenger and XLPE conductor
4	Test for thickness of insulation of XLPE conductor
5	Tensile Strength and Elongation at Break Test for both messenger and XLPE conductor
6	Hot Set Test for XLPE insulation.
7	Insulation resistance test for XLPE conductor
8	High Voltage Test for XLPE conductor

Overhead Conductors AAAC/ACSR

Standards Applicable:-	
IS: 398 (Part I & II) : 1996	Specification for Aluminum Conductors with Steel Re-inforced.
IS: 4826 : 1968	Coating of the galvanized steel wires shall be applied by the hot process or electrolysis process
IS: 398 (Part-IV): 1994	Specification for All Aluminum Alloy Conductor (AAAC)
IS: 1778 : 1980	Packaging of Overhead conductor in Wooden Drum
Proposed Tests as per IS	
1	Measurement of lay ratio.
2	Measurement of diameters of individual wire
3	Measurement of resistance of individual wire.
4	Breaking load test of individual wire.
5	Elongation test of individual wire.

Porcelain Insulators

Standards Applicable:-	
IS 1445 : 1977	Porcelain insulators for overhead power lines with a nominal voltage up to and including 1000 V
IS 2486 (Part 1) : 1993	Metal fittings of insulators for overhead power lines with nominal voltage greater than 1000 V: Part 1 General requirements and tests
IS 2486 (Part 2) : 1989	Insulator fittings for overhead power lines with nominal voltage greater than 1 000 V: Part 2 Dimensional requirements
IS 2486 (Part 4) : 1989	Insulator Fittings for Overhead Power Lines with a Nominal Voltage Greater than 1 000 V - Part IV : Tests for Locking Devices
Proposed Tests as per IS	
1	Resistance Test on conductor
2	Test for thickness of insulation and sheath
3	Tensile strength and elongation at break test for insulation & outer sheath.
4	Hot set test for insulation.
5	Insulation resistance (Volume resistivity) Test
6	High voltage test

The above list of tests is only indicative in nature and if the Employer feels, it can add some tests based on latest IS Standards. If the Employer faces any issues with respect to actual performance of a material then it can issue a notice to the Contractor for testing of that material at its sole discretion.

4.2.6.3. The Contractor shall ensure receipt of material at site within 21 days from date of receipt of dispatch instructions. In case materials are not received within 21 days from date of issue of dispatch instruction, the dispatch instruction shall stand cancelled. All expenditure incurred by Employer in performance of dispatch instruction shall be recovered from turnkey Contractor. A fresh pre – dispatch inspection would be required to issue a dispatch instruction for supplying the same lot at the site.

4.2.6.4. The turnkey Contractor shall ensure that pre-dispatch inspection for materials are intimated only when the material is completely ready for inspection. On due date of inspection, if it is found that materials are not ready in required quantities or the inspection could not be carried out due to non-availability of requisite calibrated certificate of instruments with manufacturer, closing of works on scheduled date of inspection, non-availability of sufficient testing/material handling staff at manufacturer works etc, all expenditures incurred on deployment of various inspecting officials along with a fine of Rs 50,000/- shall be recovered from the bills of the agency and re-inspection shall be carried out on expense of the Contractor. 2nd such situation at same manufacturer/supplier shall result in rejection of name of manufacturer from list of approved vendors/sub-vendors. In case sub-standard materials (old component, re-cycled materials, re-used core material, re-used transformer coil material etc) offered for inspection and are noticed during the inspection, materials shall be rejected and approval of sub-vendor shall also be cancelled for all projects.

4.3. Implications for not meeting quality requirements by Contractor

4.3.1. In case of failures in testing:

- a) The turnkey Contractor shall ensure that pre-dispatch inspection for materials is intimated only when the material is completely ready for inspection. On due date of inspection, if it is found that materials are not ready in required quantities or the inspection could not be carried out due to non-availability of requisite calibrated certificate of instruments with manufacturer, closing of works on scheduled date of inspection, non-availability of sufficient testing/material handling staff at manufacturer works etc, all expenditures incurred on deployment of various inspecting officials along with a fine of Rs 50,000/- inclusive of GST shall be recovered from the bills of the agency and re-inspection shall be carried out on expense of Contractor.
- b) 2nd such situation at same manufacturer/supplier shall result in rejection of name of manufacturer from list of approved vendors/sub-vendors. In case sub-standard materials (old component, re-cycled materials, re-used core material, re-used transformer coil material etc) offered for inspection and are noticed during the inspection, materials shall be rejected and approval of sub-vendor shall also be cancelled for all projects.
- c) In case, a material fails the pre-dispatch inspection as per GCC Clause 23, and also fails the

subsequent repeat inspection of the rectified/replaced material, the complete lot of material under inspection will be required to be replaced by the manufacturer/supplier. If in subsequent inspection of the new lot, the material again fails the inspection, then materials shall be rejected and approval of vendor/sub-vendor shall also be cancelled for all projects. In such scenario any subsequent delay in contract performance due to failure of materials in the test laboratory would be on account of Contractor and no time extension would be provided by the Employer in this regard. Any LD levies in this regard would be borne by the Contractor.

- d) Apart from the above, in case of default by vendors/manufacturers, Contractor shall also be penalized based on the no. of materials/lots get rejected as per below table:

Sr. No.	No. of Material/lot rejected in a project/district	% Penalty imposed on contract price
1	>5	5.0%
2	>3	2.5%
3	>1	1.0%

4.3.2. In case of defects found during Field inspection: There are three categories of defects found in field inspection as defined below:

- a) **CRITICAL DEFECTS:** These defects must be rectified before charging. Critical defects are those which endanger life and property. Dangerous deficiencies on safety, ground clearances, equipment earthing and protection would come this category. These are defects in presence of which the Electrical Inspector would not allow charging of the electrical installation. That is, if equipment are already energized, it should be de-energized and rectified without delay. If critical equipment like distribution transformer HT and LT line have been installed dangerously, the defect type would fall under critical category of defect. *Example : LA is not connected , DT neutral earth is missing , Earth electrodes not installed, Ground clearance not as per IE rule, Oil level low in transformer etc.*
- b) **MAJOR DEFECTS:** These defects must be rectified before operational handover (to Operation and Maintenance wing). These are major deviations from drawing and specification. These are serious deviation with respect to contract. The electrical installation can be charged temporarily. However, the defects should preferably be rectified before charging. *Example : Pole not pitched at proper depth, Brick-bats/ foundation inadequate, use of undersized earth wire, precariously loose electrical connections and mechanical fitting.,*
- c) **MINOR DEFECTS:** These defects are very minor in nature. Such defects in electrical installations keep surfacing during operation and maintenance. The installation may be charged with these defects. However they must be rectified *Example: Danger board not proper, energy meter not installed before contractual handover (before final payment is released and contract is closed), missing barbed wire, stay wire loose, loose fasteners, vegetation too close to HT/Lt line.*

Note:

1. *These defects are broad in nature. Actual field defects need to be defined more accurately by inspectors.*
2. *All pictures depicting defects should be numbered. Their number mentioned in the report shown in the table*
3. *An infrastructure schematic (single line diagram) showing DTRs, HT and LT poles duly numbered by the inspector shall be submitted along with the report. Their number shall be used to describe location of defects to be rectified subsequently.*

The corresponding penalited to be imposed has been captured below:

Sr. No.	Defect criteria	% Penalty imposed of contract cost
1	Critical Defects	1.0%
2	Major Defects	0.5%
3	Minor Defects	0% if rectified within 30 days

Annexure-A

Checklist for Quality Assurance**Distribution Transformer Substation**

S. No	Description	Status (Yes/No)	Observations	Location	Picture No.
1	Record capacity of DTR transformer used				
2	Record S. No., make and year of manufacturing of DTR transformer				
3	Safe and adequate access to distribution transformer (DTR) substation				
4	Availability of approved survey report				
5	Proper load survey is performed of the locality for perspective consumers while deciding capacity and location of DTR				
6	Expected loading of transformer using 5 years growth is performed in survey report				
7	Proper alignment of substation structure with 11 KV line				
8	Record type of poles/support used for DTR substation				
9	Record type of foundation used				
10	Proper muffing is provided on steel supports of DTR substation				
11	If DTR substation is in water logging area, its foundation is grouted in cement concrete				
12	Proper verticality of substation supports				
13	Proper pole to pole distance of substation supports.				
14	Proper erection of jumpers and connection to DTR transformers without any bent				
15	Proper binding of insulators				
16	Stay plates are properly grouted in cement concrete mixture to support DTR substation structure (if erected)				
17	Proper tensioning is there on stay set				
18	Proper alignment of stay wire with overhead conductor				

S. No	Description	Status (Yes/No)	Observations	Location	Picture No.
19	Proper erection of stay clamp using 12 mm dia nuts and bolts				
20	Proper galvanization of stay wire				
21	Thimble is provided on turn buckle of stay set				
22	Stay set installation is provided with guy insulator				
23	Proper phase to phase and phase to ground clearances maintained on the substation jumpers				
24	Steel overhead structure is properly earthed using 8 SWG wire/G.I. flat?				
25	Each 11 kV overhead equipment including transformer are individually earthed using 8 SWG Earth wire/ GI flat				
26	Danger plate is installed at appropriate height using proper size clamp. Record type and size of clamp				
27	Proper anti-climbing device (barbed wire/spike) installed at appropriate height on individual support. Record quality of wrapping of barbed wire				
28	Substation is numbered				
29	Individual substation pole is imposed/painted with the name of scheme				
30	Surface of the PCC poles is finished and there are no steel wire visible				
31	No physical damages appeared on PCC pole surface				
32	GI flat to GI flat connection using at least 2 sets of GI nut bolts and washers				
33	8 SWG GI wire/GI Flat is properly dressed with support				
34	GI wire to GI wire jointing is provided using 12 SWG GI nut bolts and washers				

S. No	Description	Status (Yes/No)	Observations	Location	Picture No.
35	GI wire connection to earth pit is using GI nut bolt and washer				
36	GI earth pipe is properly inserted inside earth without hammering				
37	Number of earth pit used for substation earthing.				
38	Pit to pit distance in meters. Is it adequate?				
39	Masonry enclosure is provided over individual earth pits				
40	Funnel is provided over earth pit				
41	Proper jumpering using binding practices/PG clamp				
42	Proper clearances to avoid bird fault on conductors of substation supports				
43	Type and size of overhead conductors used in the substation				
44	Cement-concrete grouting foundation of substation supports				
45	Measure quantum of cement concreting in any one sample support				
46	Measure cement concreting foundation in any one sample of stay set pit				
47	Proper painting/galvanizing done on steel structure				
48	Any sign of rusting found on substation structure/hardware				
49	Any broken insulator found in the substation				
50	Disc Insulators installed precariously (loose bolts/ missing cotter pins)				
51	Separate individual earth connection using GI wire/GI flat is used for neutral earthing with separate pit				
52	Dedicated transformer body earthing using GI wire/GI flat				

S. No	Description	Status (Yes/No)	Observations	Location	Picture No.
53	Bimetallic clamps are provided on 11 kV bushing				
54	No gap between busing seat and bimetallic clamp on LT as well as HT bushing while connecting conductor/cable				
55	Proper lugs are provided on termination of cables				
56	Oil is filled in cup of silica gel breather				
57	Silica gel is blue in colour				
58	Oil control valves are open between transformer tank and breather (wherever used)				
59	Oil leakage from the body/gasket of transformer and from conservator tank				
60	Record level of oil in conservator tank				
61	Transformer installed precariously (Nut / bolts / side bracing missing)				
62	Transformer is fitted with 12 mm dia nut bolts on its base channel				
63	Transformer belting is provided				
64	Dimension of transformer base channel				
65	Individual lightening arrestor are earthed with dedicated separate earth pit				
66	LA jumper connections is missing/ not proper				
67	LA is charged/ installed but not meggared				
68	Isolators/AB switch are properly aligned and its operation is smooth				
69	Operating handle (not missing eye bolt) of isolator/AB switch is earthed using flexible cable				
70	No joint in between entire length of operating pipe of isolator/AB switch				
71	Guiding hook is provided for isolator pipe movement				

S. No	Description	Status (Yes/No)	Observations	Location	Picture No.
72	Alignment of male and female contacts of isolators/AB switch and no spark during normal use				
73	Proper fuse wire is used in DO fuse/HG fuse				
74	Arching Horn is missing/ not aligned / not proper				
75	Proper size of LT cable are used between transformer and LTDB/SMC LTDB/SMC LTDB/SMC LTDB				
76	lockability and proper closing of door of LTDB/SMC LTDB				
77	Gland plate and glands are used for cable entry in LTDB/SMC LTDB				
76	No unused holes on gland plates				
77	Availability of LTDB/SMC LTDB equipment as per approved drawing and scope of work like isolator, fuse, switch, bus bar, MCCB, MCB etc.				
78	Installation of DTR as per BIS specification				
79	LTDB/SMC LTDB earthing at different points using 8 SWG GI wire				
80	Proper painting and No physical damages on LTDB/SMC LTDB				
81	Suitable loop length of cables in LTDB/SMC LTDB				
82	3 Nos earthing pit and earth mat /risers using 50X6mm GI Flat are used as under:				
a	Earth Pit – 1 for Transformer Neutral,				
b	Earth pit - 2 for Lightening Arrester,				
c	Earth pit – 3 for Equipment body earthing				
83	Deleted				
84	Deleted				
85	Quality of painting/galvanizing on substation structure				
86	DTR is newly supplied				

S. No	Description	Status (Yes/No)	Observations	Location	Picture No.
87	PG Clamps are used (wherever needed as per drwg- Jumper etc)				
88	Deleted				
89	Earthing Electrodes short/missing				
90	Commissioning Defect: DT charged/installed but not merged				
91	Fasterers (Nuts/ Bolt/ Clamps /Connector) size not as per drawing /specification				
92	Fasteners (Nuts / bolts/ Clamps / connectors) in precarious state				
93	Poles not erected properly (inadequate or missing brick bat/ foundation)				
94	Stay installation is not proper : guy insulator missing ;inadequate depth				
95	Earthing wire diameter undersize				
96	Danger plate missing/improper				
97	Earthing wire not secured / not dressed				
98	Barbed wire missing/improper				
99	DTR ground electrodes far too close				
100	Earth pit to earth pit clearance not maintained				
101	HT Fuse not provided				
102	HT fuse unit jumpering not connected properly				
103	MCCB of lower rating than specified in LOA				
104	MCCB not installed				
105	Inferior quality of Distribution Board used (makeshift, locally fabricated DBs)				
LT Line					
S. No	Description	Status (Yes/No)	Observations	Location	Picture No.
1	Availability of approved survey report with Single line diagram				
2	Correct alignment of LT line				
3	Type of poles used as per scope of the work				

S. No	Description	Status (Yes/No)	Observations	Location	Picture No.
4	Type of foundation used as per scope of work				
5	If line is passing through water logging area and its foundation is grouted in cement concrete				
6	Proper verticality of poles				
7	Any deflecting tension on LT pin insulator				
8	Proper tensioning of overhead conductor/LT cable/ABC Cable				
9	Any knot/wrapping of overhead conductor /LT cable /ABC Cable is there during erection				
10	Proper binding of insulators cable both / tension work is done				
11	Stay plates are properly grouted in cement concrete mixture				
12	Proper tensioning is there on stay set				
13	Proper alignment of Stay wire and stay set with overhead conductor is there to nullify tension				
14	Proper erection of stay clamp using 12 mm dia nuts and bolts and 50x6 mm (or more) size clamp				
15	If every 6th pole in a section of line is provided with stay sets to avoid line deflection				
16	Proper galvanization of stay wire/stay set				
17	Thimble is provided on turn buckle of stay set				
18	Proper phase to phase clearances are maintained on the line				
19	Steel overhead structure is properly earthed using 8 SWG wire				
20	Each LT pole individually earthed using 8 SWG Earth wire and separate Earth pit/Earthing coil/Earth spike				

S. No	Description	Status (Yes/No)	Observations	Location	Picture No.
21	Quality and size of danger plates is as per scope of work				
22	Danger plate is installed at appropriate height using proper clamp as per scope of work				
23	Anti-climbing device (barbed wire/spike) are installed at appropriate height on individual support				
24	Individual pole is numbered				
25	Individual pole is imposed/painted with the name of scheme				
26	Surface of the PCC poles is finished and there are no steel wire visible				
27	No physical damages appeared on PCC pole surface				
28	Cradle guard earthing is provided on each road crossing or on each LT line crossing				
29	Proper tensioning of the cradle guard wires				
30	Separate earthing on both the sides of road/line for cradle guarding are there				
31	8 SWG G.I. wire is properly dressed with support for V-Cross arm/Channel/Top clamp earthing				
32	GI wire to GI wire jointing is provided using 12 SWG GI nut bolts and washers				
33	GI wire connection to earth pit is using 12 mm GI nut bolt and washer				
34	Earth pipe is properly inserted inside earth without pipe hammering				
35	Masonry enclosure is provided over individual pipe earth pits				
36	Funnel is provided over pipe earth pit				
37	Jumpering using best binding practices/PG clamp				

S. No	Description	Status (Yes/No)	Observations	Location	Picture No.
38	Proper conductor clearances to ground is there to avoid bird fault on end sectionizer support where disc insulator are used				
39	Average pole to pole span length in the line. It should not be less than 50 m.				
40	If Pole to pole span is less than 50 m, record the reason with pole numbers				
41	Number of poles used per kilometre of the line				
42	Type and size of overhead conductors/ABC cable used in the line				
43	Shuttering is used during casting of cement concrete foundation				
44	Cement-concrete grouting foundation of end supports				
45	Quantum of cement concreting in any one sample support				
46	Cement concreting foundation in any one sample of stay set pit				
47	Proper painting is done on steel structure				
48	Any broken insulator found in the line				
49	Surface finish of painting on Steel tubular pole/RSJ/H Pole/Rail pole about 2 m from bottom and above 2 m				
50	Possible damage on ABC cable surface				
51	Piercing connections are used to take-off connection from ABC cable				
52	Muffing is used in steel steel tubular poles, rail pole, RS joint/H beam Supports				
53	Adequate tree cutting on either side of line done				
54	Pole to pole schedule enclosed with profarma				
11 KV Line					
S. No	Description	Status (Yes/No)	Observations	Location	Picture No.

S. No	Description	Status (Yes/No)	Observations	Location	Picture No.
1	Availability of approved survey report with single line diagram				
2	Correct alignment of 11 kV line				
3	Type of poles used as per scope of the work				
4	Type of foundation used as per scope of work				
5	Record whether line is passing through water logging area and its foundation is grouted in cement concrete				
6	Proper verticality of poles				
7	Cross-bracing on Double poles are provided				
8	Conductors are passing through the top groove of the insulator (creepage distance compromised)				
9	More than one joint in one span				
10	Any deflecting tension on 11 KV pin insulator				
11	Proper tensioning of overhead conductor				
12	Any knot/wrapping of overhead conductor is there during erection				
13	Proper binding of insulators is done				
14	Stay plates are properly grouted in cement concrete mixture				
15	Proper tensioning is there on stay set				
16	Proper alignment of Stay wire with overhead conductor is there to nullify tension				
17	Guy insulator, anchor plate/ thimble/ hardware are provided with stay set				
18	Proper erection of stay clamp using 12 mm dia nuts and bolts and 50x6 mm (or more) size clamp				
19	If every 6th pole in a section of line is provided with stay sets to avoid line deflection				
20	Proper galvanization of stay wire and stay set				
21	Thimble is provided on turn buckle of stay set				
22	Proper phase to phase clearances are maintained on the line				
23	Steel overhead structure is properly earthed using 8 SWG wire				

S. No	Description	Status (Yes/No)	Observations	Location	Picture No.
24	Each 11 kV pole individually earthed using 8 SWG Earth wire and separate Earth pit/Earthing coil/Earth spike				
25	Quality and size of danger plates is as per scope of work				
26	Danger plate is installed at appropriate height using proper clamp as per scope of work				
27	Anti-climbing device (barbed wire/spike) are installed at appropriate height on individual support				
28	Individual pole is numbered				
29	Individual pole is imposed/painted with the name of scheme				
30	Surface of the PCC poles is finished and there are no steel wire visible				
31	No physical damages appeared on PCC pole surface				
32	Cradle guard earthing is provided on each road crossing or on each LT line crossing				
33	Proper tensioning of the cradle guard wires				
34	Proper Guard wires are provided in case of Road crossing as per drawing specification				
35	8 SWG G.I. wire is properly dressed with support for V-Cross arm/Channel/Top clamp earthing				
36	GI wire to GI wire jointing is provided using 12 SWG GI nut bolts and washers				
37	GI wire connection to earth pit is using 12 mm GI nut bolt and washer				
38	Earth pipe is properly inserted inside earth without pipe hammering				
39	Masonry enclosure is provided over individual pipe earth pits				
40	Funnel is provided over pipe earth pit				
41	Proper jumpering using binding practices/PC clamp				

S. No	Description	Status (Yes/No)	Observations	Location	Picture No.
42	If under sized conductor used				
43	Proper conductor clearances to ground is there to avoid bird fault on end sectionizer support where disc insulator are used				
44	Proper pole to pole span length in the line. It should not be less than 50 m.				
45	If Pole to pole span is less than 50 m, record the reason with pole numbers				
46	Number of poles used per kilometre of the line				
47	Record type and size of overhead conductors used in the line				
48	Shuttering is used during casting of cement concrete foundation				
49	Cement-concrete grouting foundation of end supports				
50	Measure quantum of cement concreting in any one sample support				
51	Measure cement concreting foundation in any one sample of stay set pit				
52	Proper painting is done on steel structure				
53	Disc Insulators are installed precariously (loose bolts/ missing cotter pins)				
54	D -shaped loop for jumpers are maintained				
55	Any broken insulator found in the line				
56	Surface finish and painting on Steel tubular pole/RSJ/H Pole/Rail pole				
57	Adequate tree cutting on either side of line done				
58	Pole to pole schedule enclosed with proforma				
59	Pole numbering with inscription not done properly)				
60	Engraving of poles (Name of Manufacturer, SL Nos etc.) not done				
61	Line Spacers not used				
62	Guy insulator not used in stay wire				
63	Inadequate length of barbed wire				

Checklist for inspection of REDB (Substation)

S.N.	Description	Status (Yes/No)	Observation	Picture No.	Location as per SLD
1	Major Materials (CT/PT/CB/X'mer/Battery/ Panels /Structures/Conductor) as per specifications				
2	Record S. No., make and year of manufacturing of Power transformer				
3	Major Materials dispatched without inspection				
4	Construction as per Approved Drawing				
5	Civil works FQP documentation maintained during construction				
6	Equipment (name it) provided in the BOQ/ drawing but not installed				
7	Verification of pre-commissioning and commissioning testes of substation equipment i.e. Circuit Breaker, CT, PT, transformer, Charger, Battery, Relays, Control Panels, Switchgear, 11 KV cable etc				
8	Present condition of main equipment				
9	Functional Status of Transformer: WTI, OTI etc, Relays, Battery Charger, Battery, CB, CT, PT, Energy Meter, Control & Relay panel				
10	Transformer oil tested				
11	Transformer Relays, CT, PT , CB , Switchgears, battery sets, etc charged after test				
12	Equipment charged after commissioning test				
13	Gravel size proper				
14	Earthing of main equipment, fence etc done properly				
15	Sub Station fencing provided				

S.N.	Description	Status (Yes/No)	Observation	Picture No.	Location as per SLD
16	Cable trench made with cable trays – or cables lying on trench floor				
17	Whether Cable trenches have suitable slope to ensure automatic draining of rainwater				
18	Proper storage of equipment				
19	Cables tied on cable trays				
20	Glands, lugs used (wherever need - at cable entries)				
21	Dead end marking for cables is done				
22	Earth mat provided				
23	Undersized conductor/ cables used				
24	Correct size of earthing conductor - flats, GI wires etc used				
25	Acid proof floor used in battery room				
26	Fasteners (nut, bolts, clamps connectors, hardware etc) as per specification				
27	Switchgear rubber mats, chequer plates not provided				
28	FQP for material receipt and storage maintained by Employer				
29	Name of Feeder on Control Panel.				
30	Name of Feeder on Outgoing DP structure				
31	Working platform on 33 KV and 11 KV outdoor VCB				
32	Name of Substation board on the entrances				
33	Painting of control room, water supply position in Substation				
34	General sanitation arrangement in the control room building				
35	Internal Lighting in the substation control room				
36	Closed fencing of the substation yard				
37	Approach road to Power Transformer foundation				